

Advanced (Habanero)**Q1.** Evaluate the expression $2 \cdot 3 + 12 \div 4$.

- (A) 9
- (B) 10
- (C) 12
- (D) 15
- (E) 18

Q2. Which of the following is equivalent to $3x + 2x$?

- (A) $5x$
- (B) $6x$
- (C) $x + 2$
- (D) $2x - 3$
- (E) $x - 1$

Q3. What is the value of x in the equation $2x + 5 = 11$?

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q4. Simplify the expression $7 - 3 + 2 \cdot 4$.

- (A) 10
- (B) 12
- (C) 14
- (D) 15
- (E) 18

Q5. Which verbal phrase translates to $2x - 5$?

- (A) Two times a number minus five
- (B) Two more than a number times five
- (C) Five less than two times a number
- (D) A number minus two times five
- (E) Five minus two times a number

Q6. Evaluate the expression $(8 - 3) \cdot 2 + 1$.

- (A) 10
- (B) 12
- (C) 14
- (D) 15
- (E) 11

Q7. Simplify the expression $4x + 2x - 3x$.

- (A) $3x$
- (B) $2x$
- (C) x
- (D) 0
- (E) 3

Q8. Substitute $x = 2$ into the expression $x^2 + 3x - 4$ and evaluate.

- (A) 6
- (B) 10
- (C) 12
- (D) 14
- (E) 6

Open Ended Questions (FRQ)**Q9.** Simplify the expression $3(2x - 1) + 2(x + 4)$ and then evaluate when $x = 2$. Show all steps.**Q10.** Write an algebraic expression for the phrase 'five more than three times a number' and then evaluate when the number is 4. Show all steps.**Chef's Tips**

- Read each question carefully.
- Use the correct order of operations when evaluating expressions.
- Show all work and simplify expressions when possible.